

PORTFOLIO PROJECT REPORT

HOSPITAL EMERGENCY ROOM ANALYSIS DASHBOARD

Power BI Dashboard Project | Data Analyst Portfolio

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Executive Summary

This report documents a multi-page Power BI dashboard built to monitor Hospital Emergency Room operations. The dashboard tracks 9,216 patient visits across a 19-month window (April 2023 – October 2024), with a Consolidated View for the full period, a filterable Monthly View, a searchable Patient Details table, and a narrative Key Takeaways page.

Across the full period, patients waited an average of 35.3 minutes, and just under 60% were seen within the 30-minute target. Admissions were almost evenly split (50.04% admitted vs. 49.96% not admitted), and General Practice and Orthopedics were the leading referral destinations among patients who needed one. Saturday and Thursday were the two busiest days for patient volume.

This project demonstrates practical Power BI competencies — multi-page report design, drill-down filtering (Month/Year and date-range slicers), KPI card design, categorical and time-based visualizations, and a written narrative summary layer — suited to entry-level Data Analyst and healthcare analytics roles.

1. Project Overview

This project is an interactive Power BI dashboard built to help hospital operations staff monitor Emergency Room activity: patient volume, wait times, satisfaction, admission outcomes, and demographic and referral patterns. It is organized into three navigable pages — Consolidated View, Patient Details, and Key Takeaways — with the Consolidated View additionally filterable down to a single month.

Objective

- Track headline ER performance metrics — patient volume, average wait time, satisfaction score, and referral count — at both a monthly and multi-year level.
- Break down patient volume by admission status, age group, gender, race, and department referral to surface operational and demographic patterns.
- Identify peak demand periods by day of week and hour band to support staffing and resource planning.
- Provide a row-level Patient Details table for case-level lookup alongside the aggregated views.

Tools & Skills Used

- **Core platform:** Power BI Desktop
- **Visualizations:** KPI cards with embedded sparkline/trend visuals
- Donut charts, horizontal and vertical bar charts, and a color-scaled matrix (heatmap) visual
- Date-range and Month/Year slicers for interactive, multi-granularity filtering
- Multi-page report navigation with custom button-based menus (Consolidated View, Patient Details, Key Takeaways)
- **Business Communication:** A written narrative ('Key Takeaways') page synthesizing the visual findings into plain-language summary text
- **Data Modeling:** Row-level detail table design (Patient Details) for case-level transparency alongside aggregate KPIs

2. Dashboard Snapshot

The dashboard spans four functional views. Each is shown below at full size.

2.1 Consolidated View (Full Period: Apr 2023 – Oct 2024)

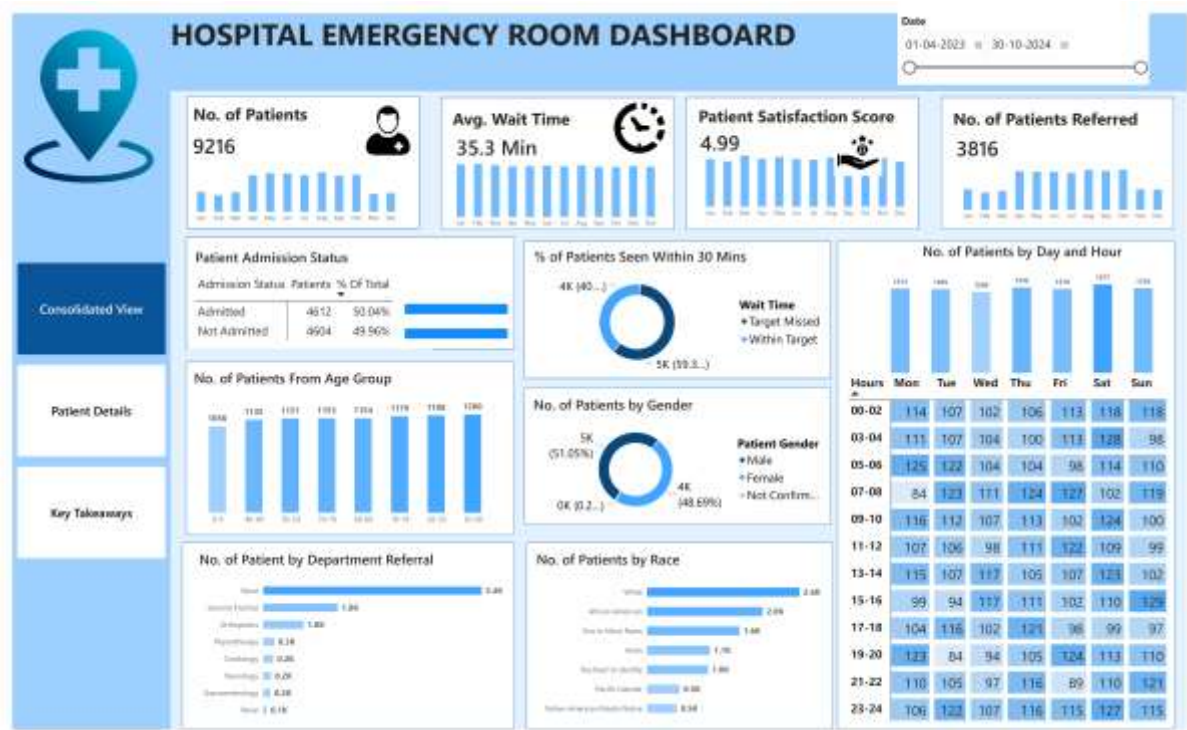


Figure 1: Consolidated View — full 19-month dataset, 9,216 patients.

2.2 Monthly View (Example: July 2023)

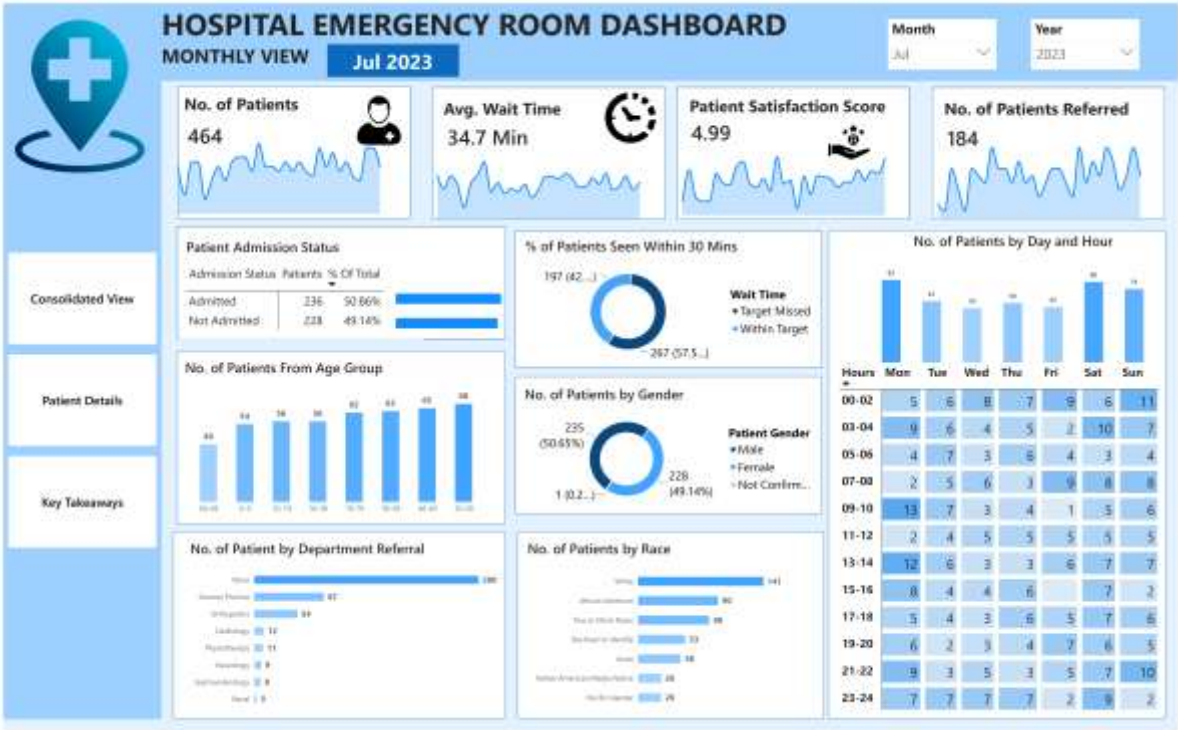


Figure 2: The same report page filtered to a single month via the Month/Year slicer, showing 464 patients for July 2023.

2.3 Patient Details



Consolidated View

Patient Details

Key Takeaways

HOSPITAL EMERGENCY ROOM DASHBOARD

Patient Details

Date: 01-04-2023 - 30-10-2024

Patient ID	Patient's Name	Patient Gender	Patient Age	Patient Admit Date	Patient Race	Patient Waittime	Department Referral	Admission Status
100-04-3990	M St Ange	Female	29	04 April 2023	White	18	None	Not Admitted
100-37-5080	V Flicker	Male	57	14 January 2024	African American	68	None	Not Admitted
100-21-0648	W Maman	Female	39	17 January 2024	Pacific Islander	22	None	Admitted
100-34-6753	B Paulus	Male	43	15 May 2024	Pacific Islander	25	General Practice	Admitted
100-34-9587	U Lamburn	Male	20	01 April 2024	Declined to Identify	24	Neurology	Not Admitted
100-40-2709	O Cammick	Male	77	08 May 2024	White	48	None	Not Admitted
100-86-0896	I Prickett	Male	2	26 March 2024	African American	23	Orthopedics	Admitted
100-66-8222	F Mullane	Female	85	23 December 2023	Asian	17	General Practice	Not Admitted
100-67-1276	S Holbird	Male	55	03 November 2023	White	11	Orthopedics	Admitted
100-70-0071	K Dearshem	Male	38	14 January 2024	African American	57	None	Not Admitted
100-72-5785	N Oding	Female	60	19 June 2024	African American	45	None	Admitted
100-74-1843	M Hallard	Female	3	17 September 2023	White	14	None	Admitted
100-74-5636	A Wernicker	Female	47	13 August 2024	Declined to Identify	25	None	Not Admitted
100-75-0109	P Ulyatt	Female	19	27 February 2024	Pacific Islander	36	Neurology	Admitted
100-81-9769	Y Monosoter	Male	28	09 March 2024	White	59	None	Admitted
100-84-7203	K Ybarra	Female	37	13 June 2023	White	55	None	Not Admitted
101-08-6796	L Wilosen	Female	72	31 July 2024	Two or More Races	48	None	Admitted
101-11-4803	V Gerswidge	Female	30	25 April 2023	Declined to Identify	38	None	Admitted
101-35-3930	W Andsworth	Female	30	24 January 2024	White	25	None	Admitted
101-35-7039	T Gaster	Male	26	08 February 2024	White	56	General Practice	Admitted
101-38-9868	J Sockington	Male	34	08 October 2024	White	28	None	Admitted
101-50-7359	D Sabey	Female	50	13 October 2023	White	56	None	Admitted
101-56-3339	M Kiduff	Female	44	08 October 2023	Asian	55	None	Admitted
101-60-5999	G Georger	Female	45	04 August 2024	White	43	None	Not Admitted
101-63-3608	P Quast	Male	38	29 July 2023	Two or More Races	53	None	Not Admitted

Figure 3: Row-level Patient Details table with Patient ID, demographics, admit date, wait time, referral, and admission status.

2.4 Key Takeaways



Consolidated View

Patient Details

Key Takeaways

HOSPITAL EMERGENCY ROOM DASHBOARD

Patient Details

Date: 01-04-2023 - 30-10-2024

Descriptive Analysis (April 2023 – October 2024)

The Emergency Room dataset covers a 19-month period and contains 9,216 unique patients.

Patient Wait Time & Satisfaction
The average patient wait time was 35.3 minutes, indicating opportunities to improve patient flow and reduce waiting times. The average patient satisfaction score was 4.99 out of 10, suggesting moderate satisfaction and highlighting the need to improve the overall patient experience.

Departmental Referrals
A substantial 5,400 patients did not require a departmental referral. Among patients who were referred, General Practice recorded the highest number of referrals (1,840), followed by Orthopedics (995), Physiotherapy (276), and Cardiology (246). These departments represent the primary referral areas within the dataset.

Peak Busy Periods
The busiest days were Monday (1,377 patients), Saturday (1,322 patients), and Tuesday (1,318 patients). The busiest admission hours were 11 AM, 7 PM, 1 PM, and 11 PM, indicating periods when additional staffing and resource allocation may be beneficial.

Patient Demographics
Patients aged 30–39 years formed the largest age group, with 1,200 patients, followed by the 20–29 age group, with 1,188 patients. The 40–49 age group was another significant demographic segment.

Race Distribution
The largest reported racial group was White (2,571 patients), followed by African American (1,951), Multiracial (1,557), and Asian (1,060) patients. A further 1,030 patients did not report or identify their race.

Admission Patterns
The dataset shows an almost equal split between admitted and non-admitted patients, 4,612 patients were admitted, while 4,604 patients were treated and released, indicating that approximately half of the emergency room patients required admission.

Summary
Overall, the analysis indicates high patient volumes, moderate satisfaction levels, and an average wait time of 35.3 minutes. General Practice and Orthopedics were the most common referral departments, while Mondays, Saturdays, and Tuesdays experienced the highest patient volumes. The demographic distribution shows a diverse patient population, with the 30–39 age group being the largest. The nearly equal admission and release rates highlight the importance of efficient emergency room capacity and resource management. These insights can support staffing optimization, patient-flow improvements, and better allocation of hospital resources.

Figure 4: Narrative summary page translating the dashboard's visuals into a written operational summary.

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3. Key Performance Indicators

The four headline KPI cards, shown here for the full 19-month Consolidated View:

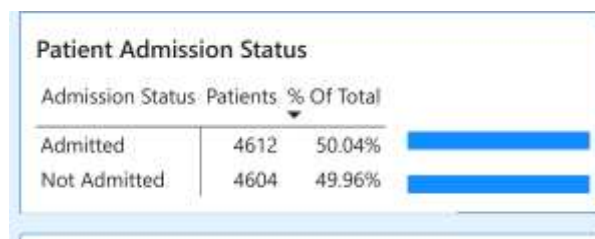
NO. OF PATIENTS	AVG. WAIT TIME	SATISFACTION SCORE	PATIENTS REFERRED
9,216	35.3 Min	4.99	3,816

Each KPI card includes a small trend visual showing month-by-month movement across the period, so a viewer can spot seasonality or drift without leaving the summary page. Average wait time (35.3 minutes) sits above a typical ER benchmark of under 30 minutes, and only 3,816 of 9,216 patients (41.4%) were referred to a specific department — the remainder were treated without a department-level referral.

4. Dashboard Components

4.1 Patient Admission Status

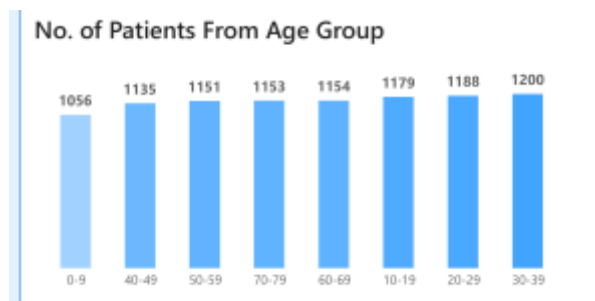
Admission Status	Patients	% of Total
Admitted	4,612	50.04%
Not Admitted	4,604	49.96%



Admission outcomes are almost perfectly split down the middle.

The near 50/50 admission split means roughly half of all ER visits over the 19-month period resulted in a hospital admission — a useful baseline for forecasting inpatient bed demand directly from ER volume.

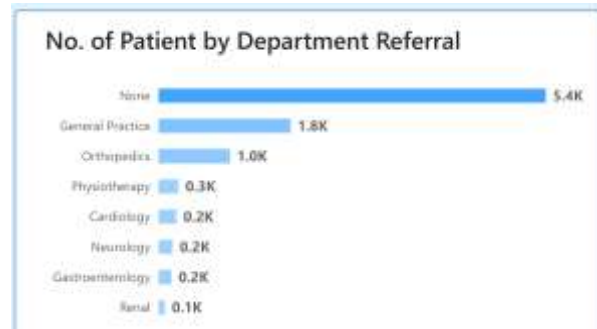
4.2 No. of Patients by Age Group



Patient volume by age band, ranked lowest to highest.

The 30–39 age group is the single largest cohort (1,200 patients), with 20–29 close behind (1,188). Volume is fairly evenly distributed across working-age bands (20s through 70s), with the 0–9 age group the smallest (1,056).

4.3 No. of Patient by Department Referral



Referral volume by destination department (None = discharged without a specialist referral).

5,400 patients (58.6% of all visits) required no department referral. Among those referred, General Practice (1,840) and Orthopedics (995) account for the large majority, with Physiotherapy, Cardiology, Neurology, Gastroenterology, and Renal each in the low hundreds or less.

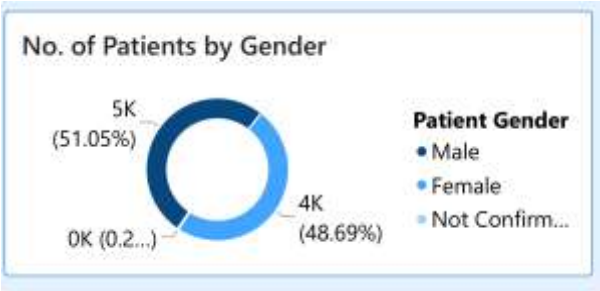
4.4 % of Patients Seen Within 30 Minutes



Wait-time performance against the 30-minute target.

59.3% of patients were seen within the 30-minute target, meaning roughly 2 in 5 patients (40.7%) waited longer — a direct, trackable service-level indicator for ER flow management.

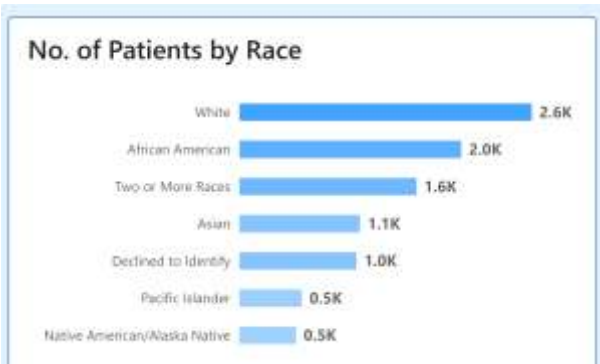
4.5 No. of Patients by Gender



Patient volume split by reported gender.

The patient population is close to evenly split between male (51.05%) and female (48.69%), with a small fraction (0.2%) not confirmed — a demographic balance worth noting for equitable service planning.

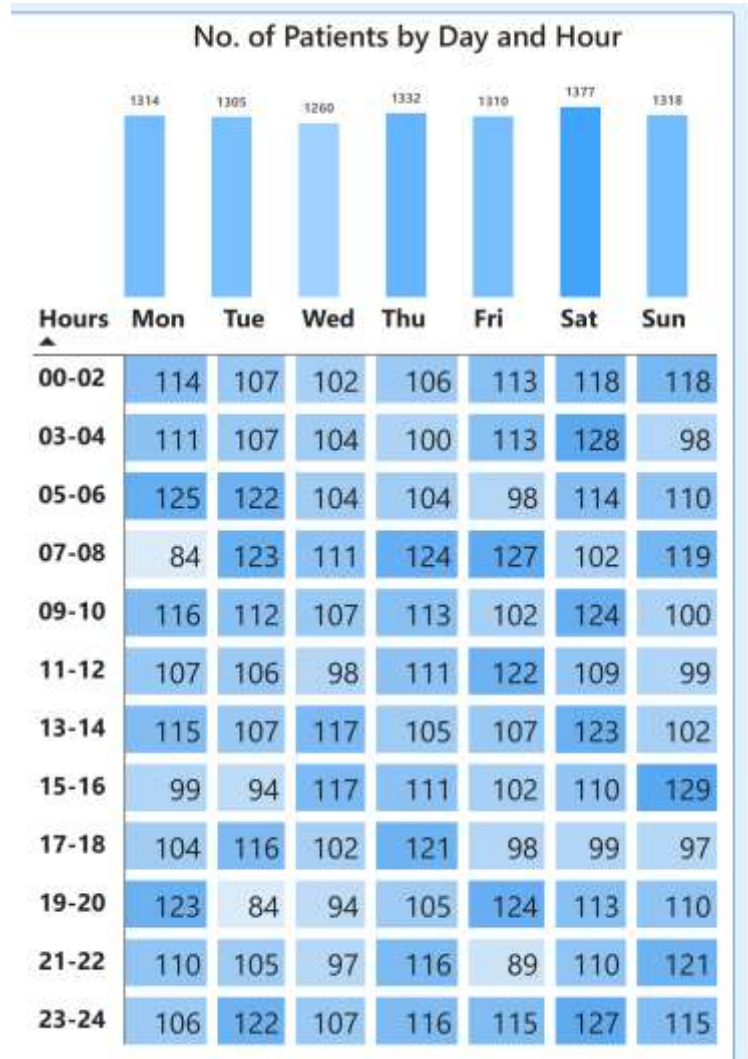
4.6 No. of Patients by Race



Patient volume by reported race/ethnicity.

White (2.6K) and African American (2.0K) patients form the two largest groups, followed by patients identifying as Two or More Races (1.6K) and Asian (1.1K). 1.0K patients declined to identify their race.

4.7 No. of Patients by Day and Hour



Color-scaled matrix of patient volume by day of week and 2-hour band, with day totals along the top.

By day-of-week total, Saturday is the busiest day (1,377 patients), followed by Thursday (1,332) and Sunday (1,318); Wednesday is the quietest (1,260). At the hour-band level, the 23:00–24:00 band carries the single highest combined total (808 patients across the week), with 07:00–08:00 close behind (790) — both useful signals for overnight and early-morning staffing decisions.

5. Key Insights & Recommendations

- Wait-time performance has room to improve: only 59.3% of patients are seen within the 30-minute target, and the 35.3-minute average sits above common ER benchmarks — worth a root-cause review of triage staffing during peak bands (23:00–24:00 and 07:00–08:00).
- Admissions run close to 50/50, giving ER volume strong predictive value for downstream inpatient bed planning — a rolling ER-volume forecast could feed directly into bed-capacity planning.
- General Practice and Orthopedics dominate specialist referrals; ensuring adequate on-call or fast-track capacity in these two departments specifically would address the majority of referral-driven wait time.
- Saturday and Thursday consistently see the highest patient volumes — proactively scheduling additional ER staff on these two days, rather than a flat weekly schedule, would better match capacity to demand.
- The near-even gender and broad age/race distribution suggest the ER serves a representative cross-section of the community, which is useful context for equitable-access reporting.

6. Challenges & Learnings

- Cross-checked the dashboard's own auto-generated "Key Takeaways" narrative against the underlying chart data and found a discrepancy: the narrative text names Monday, Saturday, and Tuesday as the busiest days, while the day-of-week totals actually shown on the dashboard rank Saturday (1,377), Thursday (1,332), and Sunday (1,318) highest. This report uses the raw charted totals rather than the auto-generated narrative, and flags the mismatch as a reminder to verify BI-tool-generated summary text against the source visuals before presenting it to stakeholders.
- Worked with mixed units across charts (raw counts on the Monthly View vs. K-rounded totals such as "5.4K" and "2.6K" on the Consolidated View) and cross-referenced the dashboard's own Key Takeaways page to recover exact figures (e.g., 5,400 and 2,571) where the chart labels were rounded.
- Balanced four KPI cards, six charts, a 12-row × 7-column heatmap, and a slicer on a single Consolidated View page while keeping every label legible at a glance.

7. Conclusion

This dashboard demonstrates the ability to design a multi-page, multi-granularity Power BI report — combining KPI monitoring, categorical and time-based visual analysis, row-level detail, and a plain-language narrative layer — for an operational healthcare use case. It reflects core data analyst skills: dashboard design, interactive filtering, data verification (including catching a narrative-vs-data inconsistency), and translating operational metrics into actionable staffing and capacity recommendations.